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LISA 2020: Promoting Statistical Practice and Collaboration in Developing Countries[1]

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2.1 Introduction

Statistics and data science collaboration laboratories ("stat labs") are collections of collaborative statisticians and data scientists who collaborate with domain experts to enable and accelerate research and make data-driven business and policy decisions (Vance and

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Pruitt 2022). In short, stat labs drive sustainable development while also building further capacity for development, which we consider to be sustainable actions that affect society positively (Vance and Love 2021).

Stat labs achieve outstanding results by using projects to train their staff and students to further build their capacity, teaching short courses and workshops to improve statistical skills and data literacy widely, and collaborating with domain experts to transform evidence into action to positively impact development (Olubusoye et al. 2021; Vance 2015b; Vance and Smith 2019). Members of stat labs can learn from each other. When a project requires specialized statistical methodology or data science skills (e.g., spatial statistics, causal analysis, Bayesian analysis, Big Data), an individual working on the project can seek advice from other stat lab members with the needed skills.

Similar to how a stat lab can upscale the impact of an individual collaborative statistician or data scientist by surrounding her with mentors and stat lab members who can complement her technical skills and teach her new ones, a network of stat labs can multiply the impact of its constituent stat labs. Research networks—and the international collaborations they stimulate—have been shown to increase the production of scientific papers and their citations (Jacob and Meek 2013; OECD 2011). Networks facilitate mentorship between network members (Vance et al. 2017a, 2017b), enable exchanges, share best practices, provide purpose and motivation to other members, and yield increased opportunities for all members.

We believe that the LISA 2020 Network is such a network. LISA 2020 is a program to build statistical analysis and data science capacity in developing countries to transform evidence into action. It does this by helping to create, strengthen, and sustain stat labs that have three missions:

1. Train statisticians and data scientists to become effective, interdisciplinary collaborators who can move between theory and practice to solve problems for real-world impact
2. Serve as research infrastructure for researchers and decision-makers to collaborate with statisticians and data scientists to enable and accelerate research and data-based decisions that make a positive impact on society
3. Teach short courses and workshops to improve statistical skills and data literacy widely.

LISA 2020 is centered at the Laboratory for Interdisciplinary Statistical Analysis (LISA) at the University of Colorado Boulder, United States. Since its creation in 2012, the goal of LISA 2020 has been to create a network of at least 20 stat labs in developing countries by 2020 (hence the name "LISA 2020"), specifically by 20 October 2020, which was the third UN-designated World Statistics Day.

The goal of this chapter is to explain the history, current state, and future directions of the LISA 2020 Network. In Section 2.2, we describe the origins of the LISA 2020 Network and key events in its history. Section 2.3, characterizes the current state of the network. In Section 2.4, we discuss the purpose of the LISA 2020 Network, how the network can improve outcomes and impacts of all its members, and how we can sustain this. We preview the future directions for the LISA 2020 Network in Section 2.5, and we conclude in Section 2.6.

2.2 The Origin and History of the LISA 2020 Program

2.2.1 The Beginning of LISA 2020

LISA 2020 was created in 2012 by Dr. Eric Vance, who was the Director of LISA at Virginia Tech in Blacksburg, Virginia, United States. The program came from three international visions for LISA he publicized in 2011 on the LISA website and in the *LISA Annual Report for 2010–2011* (Vance 2012b):

2011 Vision 1. ON-THE-GROUND STATISTICIANS: LISA graduate students will work on large, global, interdisciplinary research projects as "on-the-ground" statisticians (Seiss et al. 2014).

2011 Vision 2. STATISTICAL EXCHANGES: LISA students will learn to be internationally aware and culturally competent collaborative statisticians by participating in short-term (6-month) student exchanges within a network of statistical consulting and collaboration centers.

2011 Vision 3. CAPACITY BUILDING: Establish statistical consulting and collaboration centers at universities in countries currently lacking the infrastructure and statistical capacity to assist researchers with statistics.

Vision 1, on-the-ground statisticians, was based on lessons learned from two large, international LISA research projects. The first was a study sponsored by the World Bank of water supply and demand in rural Senegal (Hall et al. 2014b). Local enumerators asked more than 300 questions of around 2,000 households about their use of water for various domestic and economically productive purposes (e.g., growing garden crops, irrigating fields, watering livestock). The overall research question was whether investments in constructing water supply infrastructure (e.g., pumps, holding tanks, distribution pipes) would cause enough economic activity to pay for the initial investment (Hall et al. 2015a). LISA was contracted to analyze the data (Hall et al. 2015b) and found that some relevant data was missing (i.e., an important source of water [surface water] was not queried) and that the extant data was riddled with inconsistencies and ridiculous values due to enumerator error. For example, households were asked about their sales of irrigated crops; the data indicated that one farmer had, incredibly, sold one watermelon for US$50,000.

The challenge of understanding, cleaning, and analyzing data about a context foreign to LISA statisticians prompted the idea of embedding a statistician within the team of project designers to ensure that no relevant question was accidentally dropped from the survey and within the team of enumerators to flag suspicious data points, clean the data in real time, and retrain enumerators as necessary. We called this person an "On-the-Ground Statistician" and implemented this idea on a new project sponsored by the Millennium Challenge Corporation to understand the impact of installing water hand pumps in rural Mozambique (Hall et al. 2014a). The scheme worked exactly as intended! Statisticians on the ground helped redesign the study at the last minute, ensured that all relevant data was collected, verified all data points in the field within hours after they had been collected, retrained enumerators when necessary, and provided weekly updates and summary tables to the research team and funding agency. The result was a high-quality dataset that could be easily modeled and analyzed by the same on-the-ground statisticians who

now understood the local context of the data production (Seiss et al. 2014; Van Houweling et al. 2017). However, an on-the-ground statistician from the United States is expensive and infeasible for most international projects. A more feasible solution would be to embed a local statistician within the research team who already understood the local context of the research and had the necessary technical skills to guide the research team toward collecting, modeling, and analyzing high-quality data to answer relevant research, business, and policy questions.

Vision 2, statistical exchanges, was motivated by a desire to improve LISA at Virginia Tech by exposing it to perspectives from around the world so as to become *the* premier academic statistical consulting and collaboration laboratory and to provide opportunities for LISA graduate students to travel abroad. At the time, LISA had funding for four, full-time (20 hours/week) graduate students to engage in statistical collaboration. If we could find a match with a foreign university who trained and funded their graduate students to be full-time (20 hours/week) statistical collaborators, then we could make an exchange, with LISA paying its graduate student to work in the foreign stat lab and the foreign university paying their graduate student to work in LISA. As a result, LISA would learn from the experience and ways of collaborating from the foreign exchange student, and the foreign stat lab would learn about LISA's methods. Several LISA students investigated options for exchanges but never found a foreign stat lab that trained and funded graduate students to engage in statistical consulting and collaboration. All of the labs that we found employed permanent staff to engage in *consulting* rather than graduate students to engage in statistical *collaboration*.

Vision 3, capacity building, was motivated by the realization that the LISA model of training students to become effective interdisciplinary collaborators to assist researchers, businesses, and policymakers in statistics was exceptional, had great potential for impact (Vance and Pruitt 2022), and could be adapted and adopted widely for the benefit of society. Furthermore, to train future on-the-ground statisticians who fully understood the local context of projects (Vision 1) and facilitate exchanges of students between LISA and foreign stat labs (Vision 2), more capacity would have to be built. In other words, LISA would need to help establish new stat labs around the world.

In 2012, O. Olawale Awe published a paper calling for all universities in Africa to teach statistical consulting and establish statistical consulting units (Awe 2012). In his paper, Awe summarized the training methods of LISA and concluded, "There is a need for other institutions of higher learning across the continent [of Africa] to follow the same pattern."

While searching for possible exchange opportunities for his students or potential locations for creating new stat labs, Vance read Awe's paper and realized that there was, in fact, an expressed need for and demand to create stat labs in Africa modeled after LISA. A new vision was then established: to build statistical capacity and research infrastructure in developing countries.

Vance explained this new vision to Tim Howland, an officer at Virginia Tech responsible for cultivating charitable donations to the university. On March 2, 2012, Howland and Vance had a conversation that established the goals and the name of the LISA 2020 program:

Howland: "How will you build statistical capacity in developing countries?"
Vance: "By helping to create stat labs in the pattern of LISA."
Howland: "How many stat labs?"
Vance: "Twenty."

Howland: "By when?"
Vance: "Twenty by the year 2020, and we'll call the program 'LISA 2020.'"

Thus was established the program, the goal of creating 20 stat labs in the pattern of LISA by the year 2020, and the name "LISA 2020."

2.2.2 The Origins of LISA 2020

In one sense, the roots of LISA 2020 go as far back as the roots of LISA, which is to 1948 when Dr. Boyd Harshbarger created the Statistical Laboratory at Virginia Polytechnic Institute in Blacksburg, Virginia within the university's Agricultural Experiment Station. Based on the success of the Statistical Laboratory, in 1949 the Department of Statistics—offering an M.S. degree—was approved and housed in the College of Arts and Sciences (Arnold 2000). It is important to emphasize that the Statistical Laboratory predated and paved the way for the Department of Statistics.

In 1973, the Statistical Laboratory was reorganized to become the Statistical Consulting Center under the direction of Dr. Ray Meyers to expand services across the university (now named Virginia Polytechnic Institute and State University) (Arnold 2000). A major change implemented by Myers was to emphasize the training of graduate students in statistical consulting (Arnold et al. 2013). The Statistical Consulting Center was reborn in 2008 as the Laboratory for Interdisciplinary Statistical Analysis (LISA) under the leadership of interim director Laura Freeman and its first (and only) Director Dr. Eric Vance (Arnold et al. 2013). A major focus of LISA was to emphasize collaboration over consultation and to revitalize the training program of its students in interdisciplinary collaboration (Vance 2009, 2015b).

Another transition occurred in 2016 when Vance moved LISA to the University of Colorado Boulder, maintaining LISA's traditions and methods, but operating in the context of a different university and being housed within the Department of Applied Mathematics instead of a statistics department. Concurrently, the LISA 2020 program moved from Virginia Tech to the University of Colorado Boulder while maintaining its original purpose, methods, and personnel (Vance 2017).

LISA 2020 also stems from the travel experiences of Vance as a young man. For 5 years between his undergraduate education and graduate school at Duke University for his M.S. and Ph.D. in Statistical Science, Vance traveled around the world three times through 67 countries in Europe, Australia and New Zealand, South and Southeast Asia, Central and South America, and Africa. In 2002, during his travels through 17 countries in Africa, Vance attempted to hitchhike from the territory of Western Sahara to Mauritania through a part of the Sahara Desert that had no roads (Vance 2012a). He was able to find a ride on the top of a four-wheel-drive vehicle (see Figure 2.1) and rode for five hours bungee-corded to its roof rack to the border of Western Sahara territory and Mauritania, which was laden with mines at the time. The couple who drove Vance to the border felt it would be unsafe to drive through the minefield with him on top of their car.

At the border checkpoint, Vance met two Frenchmen who owned and were driving a bus south from France to sell in West Africa. They agreed to give him a ride in their bus across the minefield after the paperwork to cross the border was completed. At the border checkpoint Vance also met a Ph.D. biologist studying the Saharan desert fox, also known as the fennec fox. When the biologist learned that Vance was to be studying statistics in graduate school, he exclaimed, "A statistician! I need to talk to you." The biologist began to explain how he was struggling to count the foxes and make sense of the data he could collect about them. But before they could get very far into the details of what was Vance's



FIGURE 2.1
Photos of Vance hitching a ride on top of the vehicle (top-left), the border checkpoint (top-right), the consultation with the Ph.D.-biologist (bottom-left), and the Frenchmen's bus stuck on a sand dune in the middle of the Sahara Desert (bottom-right).

first-ever statistical consultation, the Frenchmen's bus was ready to depart, and Vance had to abruptly leave the biologist behind to complete the 4-day journey across the desert.

Vance's travels and his abbreviated consultation with the biologist are relevant to LISA 2020 because these experiences informed his motivations. Vance had great experiences traveling around the world and wanted to provide his students in LISA opportunities to travel internationally. He recognized the tremendous benefits of LISA to supply training opportunities for his students to apply statistics to solve real-world problems and for enabling and accelerating research and data-driven decisions within the Virginia Tech community. His experience at the border checkpoint taught him that there are literally Ph.D.-scientists wandering the Sahara Desert looking for statisticians to collaborate with! These experiences combined to help Vance form the idea for the LISA 2020 program because they showed him that it is not just scientists, businesses, and policymakers in the United States who need to collaborate with statisticians and data scientists. Decision makers around the globe need the option to collaborate with statisticians and data scientists, especially in developing countries.

2.2.3 The Timeline of Key Events for LISA 2020

In Table 2.1, we list key events in the history of LISA 2020, which include sources of funding, symposia and conferences, and exchange visits between stat lab personnel.

TABLE 2.1

Timeline of Key Events for LISA 2020

2012
Vance establishes the idea for the LISA 2020 program; travels to the University of Juba in South Sudan
Emanuel Msemo from Tanzania begins pursuing his M.S. in Statistics at Virginia Tech (VT)
2013
Vance is awarded a Google Research Award to establish the LISA Fellows program
O. Olawale Awe from Obafemi Awolowo University (OAU), Ile-Ife, Nigeria selected to serve as the first LISA Fellow, training at Virginia Tech for one year
iAGRI/USAID Award: "Growing Research Capacity at Sokoine University of Agriculture (SUA) by Creating a Statistical Collaboration Laboratory"
Vance visits SUA in Tanzania (also in 2015) and the Federal University of Rio Grande do Norte (also in 2014)
2014
Benedicto Kazuzuru from SUA, Tanzania, visits VT for 6 months
Richard Ngaya from Tanzania begins pursuing his M.S. in Statistics at VT
Jingli Xing from Renmin University in Beijing, China, visits VT for six months
O. Olawale Awe visits Alex Hanlon at the University of Pennsylvania School of Nursing to train with the BECCA lab
Mohammad Djedour from University of Science and Technology Houari Boumediene (Algeria) visits VT for 2 weeks
Ayele Taye Goshu from Hawassa University, Ethiopia, visits VT for 2 months
2015
VT graduate student Ian Crandell spends 6 months in Nigeria at OAU and the University of Ibadan
VT graduate student Adam Edwards travels to Tanzania to support the newly established SUALISA
Vance visits Nigeria, Tanzania, Ethiopia, India, and Brazil
2016
LISA 2020 Advisory Council formed with support from the American Statistical Association (ASA)
Olusanya Olubusoye from UI-LISA visits VT and Purdue University for 3 weeks
Vance moves LISA to the University of Colorado Boulder (CU Boulder)
Imran Khan from Kashmir, India serves as a postdoctoral research fellow to help establish LISA at CU Boulder; visits the UCLA Statistical Consulting Group for 2 weeks in 2017
Demisew Gebru from Hawassa University, Ethiopia, visits Jim Rosenberger at Pennsylvania State University for 3 months and CU Boulder for 1 week
George Uchechukwu spends 6 weeks working at UI-LISA with Olubusoye
2017
First LISA 2020 Symposium in Marrakech, Morocco in conjunction with ISI's 61st WSC with nine participants
Vance presents Keynote at the First International Conference of the Nigerian Statistical Association
2018
LISA 2020 funded by USAID Accelerating Local Potential program
Idowu Adarabioyo spends 6 weeks learning at Anchor University Laboratory for Interdisciplinary Statistical Science & Data Analysis (AULISSDA) from O. Olawale Awe
2019
Second LISA 2020 Symposium in Kuala Lumpur, Malaysia in conjunction with ISI's 62nd WSC with 34 participants
2020–2021
Third LISA 2020 Symposium held virtually with hundreds of attendees; local and regional symposium events held in Ghana (KNUST), Tanzania (Mzumbe), Nigeria (NSUK, MOUAU, OAU), South Africa (Stellenbosch), Pakistan (Lahore), Brazil (UNIOESTE), and Egypt (Alexandria)

Vance's first international travel as Director of LISA 2020 was in June 2012 to the University of Juba in South Sudan. There he met three statisticians eager to start a stat lab (see Figure 2.2), including Loro Gore, who relayed to Vance a story about a pharmacy M.Sc. student who struggled for 9 months to solve a statistics problem in his thesis. The problem was seemingly too difficult for the student to overcome, so he decided to withdraw from his studies. The Registrar at the time, Dr. Martin Baru, asked the student to meet with Gore before dropping out, and 45 minutes later the problem was solved! This anecdote highlights the potential for impact of a collaborative statistician and the need for stat labs in *all* universities so that future researchers, business people, and policymakers do not become so discouraged and frustrated with their lack of capability in statistics and data science.

The first funding support for LISA 2020 was a Google Faculty Research Award to Vance in 2013, facilitated by Dr. Steve Scott. This award created the LISA Fellows program and sponsored one statistician from a developing country to work in LISA at Virginia Tech for



FIGURE 2.2
Photos of Vance with Jaffa, Magdalena, and Gore at University of Juba (top-left); with Awe and Pruitt at VT (top-right); with Awe and Crandell at OAU (middle-left); with Edwards at SUA (middle-right); with Advisory Council (Cochrane, Love, Olubusoye, Altman, Pruitt, Morganstein, Clarke, Edwards, Cule, Rosenberger, and Pollack [not pictured]) in San Diego (bottom-right); at NSA roundtable discussion in Lagos (bottom-right).

1 year. In only 3 weeks, LISA received 108 full fellowship applications from 34 countries. The selection committee named three finalists—Dr. Carla Vivacqua, Genelyn Sarte, and O. Olawale Awe—with Awe selected as the first LISA Fellow. Several other semi-finalists have remained connected to the LISA 2020 Network and created their own stat lab; see, for example, Goshu (2016) and Amin and Vance (2016).

A follow-up Google Faculty Research Award enabled LISA graduate student Ian Crandell to travel to Nigeria for 6 months in 2015 to help Awe build the capacity of Awe's newly created Laboratory for Interdisciplinary Statistical Analysis and Collaboration (LISAC) at Obafemi Awolowo University—the first stat lab established in the LISA 2020 Network. While in Nigeria, Crandell, Vance, and Awe visited the University of Ibadan (UI) to help them establish UI-LISA, the third stat lab in the growing network in June 2015. They also visited Afe Babalola University Ado-Ekiti (ABUAD), which became a full member of the network in 2020.

Also, in 2015, a second LISA graduate student, Adam Edwards, was sponsored by the USAID-funded iAGRI program to visit the newly created Sokoine University of Agriculture Laboratory for LISA (SUALISA) in Tanzania—the second stat lab in the Network—to help build its capacity and sustainability (Msemo and Vance 2015; Vance and Magayane 2014). This visit was facilitated by previous exchanges: an educational visit by Emanuel Msemo from 2012 to 2014 to earn his M.S. in Statistics at Virginia Tech, a visit by Vance to SUA in 2013, a research visit to LISA by Dr. Benedicto Kazuzuru in 2014, and an educational visit by Richard Ngaya from 2014 to 2016 to earn his M.S. in Statistics.

In 2016 the American Statistical Association (ASA) funded Vance's proposal to convene a LISA 2020 Advisory Council. This—along with funding from the Center for Open Science—enabled Prof. Olusanya Olubusoye from UI to join other council members for a meeting at the ASA Conference on Statistical Practice in San Diego, CA. The secretariat of LISA 2020 moved from Virginia Tech to the University of Colorado Boulder in the summer of 2016 when Vance moved LISA.

In 2017, the International Statistical Institute (ISI) provided a room for the first LISA 2020 Symposium held in conjunction with the ISI's 61st World Statistics Congress in Marrakech, Morocco. This was the first opportunity for most of the nine attendees to hear directly from each other about their experience creating and running stat labs. Also, in 2017, Vance delivered a keynote at the First International Conference of the Nigeria Statistical Association in Lagos and, with Awe and Olubusoye, convened a roundtable discussion with several future LISA 2020 members. See Figure 2.2 for photos of this and other key events.

Vance was awarded funding from the United States Agency for International Development (USAID) in 2018 to accelerate local potential at higher education institutions in low- and middle- income countries to transform evidence to action to address development challenges. One of the activities this award funded was the second LISA 2020 Symposium, held in 2019 in Kuala Lumpur, Malaysia in conjunction with the ISI's 62nd World Statistics Congress. Thirty-four participants from 11 countries engaged in person over more than 3 days to share best practices and develop their collaboration and stat lab management skills. See Figure 2.3.

On World Statistics Day, 20 October 2020, during the keynote of the online LISA 2020 Symposium—which highlighted best practices in statistical collaboration and practice in developing countries from stat labs throughout the LISA 2020 Network from October 2020 to May 2021—Vance announced that the LISA 2020 program had achieved its eight-year-old goal of creating a network of at least 20 stat labs and furthermore had surpassed this goal by 40%, welcoming 28 stat labs as full members of the LISA 2020 Network. Table 2.2 shows when the LISA 2020 stat labs were established or joined the network.



FIGURE 2.3
Ten participants from seven countries attended the First LISA 2020 Symposium in Marrakech in 2017. Thirty-four participants from 11 countries attended the Second Symposium in Kuala Lumpur in 2019.

TABLE 2.2

Timeline of the Establishment of Stat Labs in the LISA 2020 Network

2014

1. O. Olawale Awe founds LISAC (Laboratory for Statistical Analysis and Collaboration) at Obafemi Awolowo University, Nigeria
2. Benedicto Kazuzuru and Emanuel Msemo found SUALISA at Sokoine University of Agriculture, Tanzania

2015

3. Olusanya Olubusoye founds UI-LISA at the University of Ibadan, Nigeria
4. Ayele Taye Goshu founds HwU SCC (Hawassa University Statistical Collaboration Center) in Ethiopia
5. Carla Vivacqua joins LEA (Laboratório de Estatística Aplicada) at Federal University of Rio Grande do Norte, Brazil to the LISA 2020 Network

2017

5. O. Olawale Awe founds AULISSDA (Anchor University Laboratory for Interdisciplinary Statistical Science & Data Analysis), Nigeria as LISAC goes dormant
6. Ayele Gebeyehu founds WUSCCC (Wolkite University Statistical Consultancy and Collaboration Center), Ethiopia

(*Continued*)

TABLE 2.2 (*Continued*)

Timeline of the Establishment of Stat Labs in the LISA 2020 Network

2018

7. Haftom Temesgen founds MULISDA (Mekelle University Laboratory for Interdisciplinary Statistical Data Analysis), Ethiopia
8. M.O. Adenomon founds NSUK-LISA at Nasarawa State University, Keffi, Nigeria
9. Kayode Ayinde and Ezra Gayawan found FUTA-LISA at the Federal University of Technology, Akure, Nigeria

2019

10. Atinuke Adebanji founds KNUST-LISA at Kwame Nkrumah University of Science and Technology, Ghana
11. Bhaswati Ganguli founds data science lab at the University of Calcutta, India
12. Joyce Otieno founds MU-LISA at Maseno University, Kenya; see Mawora et al. (2022)
13. Albert Ayorinde Abegunde, Olubola Babalola and O. Olawale Awe rebranded LISAC to become EDM-LISA OAU.

2020

14. Osama Hussien founds AU-LISA at Alexandria University, Egypt
15. Asifa Kamal founds LISA-LCWU at the Lahore College for Women University, Pakistan
16. George Uchechukwu founds MOUAU-LISA at Michael Okpara University of Agriculture, Umudike, Nigeria
17. Rita Lima founds Statistical Collaboration Laboratory (SCL) at Federal University of Piauí (UFPI), Brazil
18. Tonya Esterhuizen joins SU-Biostatistics at Stellenbosch University, South Africa to the LISA 2020 Network
19. Temidayo Apata founds FUOYE-LISA at the Federal University Oye Ekiti, Nigeria
20. Imran Khan founds SKUAST-K-LISA at Sher-e-Kashmir University of Agricultural Sciences and Technology of Kashmir, India
21. Lubna Naz founds KU-LISA at the University of Karachi, Pakistan
22. Miguel Opazo founds LEE (Laboratório de Estatística Espacial) at the Universidade Estadual do Oeste do Paraná (UNIOESTE), Brazil
23. Adarabioyo Idowu founds ABUAD-CENSAC (Centre for Statistical Analysis and Collaboration) at Afe Babalola University Ado-Ekiti (ABUAD), Nigeria
24. Hope Mbachu founds IMSU-LISA at Imo State University, Nigeria
25. Tadesse Awoke founds UoGStatLab at the University of Gondar, Ethiopia
26. Enobong Udoumoh founds FUAM-LISA at the Federal University of Agriculture, Makurdi, Nigeria
27. Justine Mbukwa founds MULISA at Mzumbe University, Tanzania
28. Daniels Akpan founds CLEDA-LISA at the African Centre for Education Development, Nigeria

2021

29. Bashiru Saeed and Ken Kubuga found TaTU DataScience Lab at Tamale Technical University, Ghana
30. Abdul-Aziz Abdul-Rahaman founds KsTU-DATALINK at Kumasi Technical University, Ghana
31. Happiness Obiora-Ilouno founds NAU-LISA at Nnamdi Azikiwe University, Nigeria
32. Oludare Ariyo founds FUNAAB-LISA at the Federal University of Agriculture, Abeokuta, Nigeria
33. Lamidi-Sarumoh Alaba founds GSU-LISA at Gombe State University, Nigeria
34. Benjamin Odoi and Christiana Nyarko found UMaT-LISA at the University of Mines and Technology, Tarkwa, Ghana
35. Muyiwa Agunbiade founds UNILAG-LISA at the University of Lagos, Nigeria

2.3 The Current State of the LISA 2020 Network

The LISA 2020 Network currently has 35 full members, 6 transitional members, and 7 proposed members (see Figure 2.4). Vance and Pruitt (2022) describe a seven-step procedure for new labs to become full members. Briefly, labs must complete a stat lab plan to become a proposed member, respond to feedback on their plan to become a transitional member,

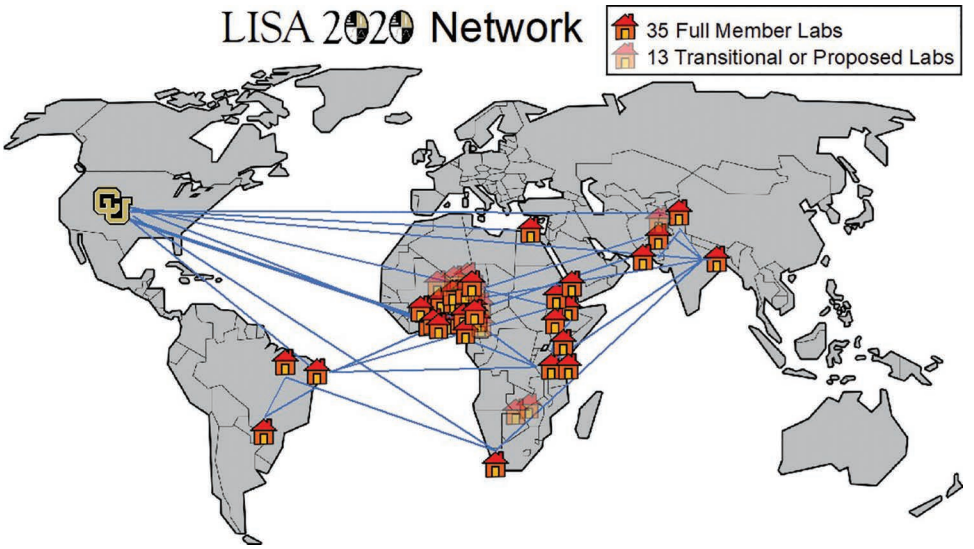


FIGURE 2.4
Map of the LISA 2020 Network's 35 full members and 13 transitional or proposed stat labs indicated by transparency.

and then finalize their plan, report metrics, present their lab to the full network, and be approved by two-thirds of the current full members to become a full member of the LISA 2020 Network.

From 2014 to 2018, the stat labs reported having trained 141 statisticians to communicate and collaborate with more than 1,000 researchers on both short-term consultations and longer term collaborations. They also reported having taught 49 short courses to improve the statistical skills of 1,396 attendees.

In 2019, Dr. Kim Love streamlined the process for stat labs to report their metrics. The network also began to disaggregate metrics referring to numbers of people by gender. This is because gender equality is an important consideration for the LISA 2020 Network. Women bring unique and important perspectives to collaborative settings, development issues, and capacity building, and in many developing countries, they face significantly greater barriers and unrealistic standards relative to their male colleagues. The directors and staff of the LISA 2020 Network stat labs are enthusiastic about gender inclusivity, and by disaggregating these metrics by sex, we remain informed of the state of gender equality in the network's activities and can recommend remedial measures if and where they are necessary.

By 2019 the network had grown to nine full members and would reach 13 full members by the end of the year. One important metric is the number of collaborative statistician trainees produced by the labs. These are individuals (faculty, staff, and students) who have statistics or data science expertise and are trained to use these skills in a collaborative setting. Individuals may be trained in several ways, including (1) participating in stat lab projects, (2) enrolling in a consulting/collaboration course in the department, (3) attending collaborative training specially organized by the lab, (4) participating in collaborative coaching and feedback sessions reviewing video-recorded meetings with domain experts, and/or (5) attending regular meetings where stat lab projects are discussed. Between 2019 and 2021, 36 stat labs have reported training 690 faculty and staff members (32.9% female) in the practice of effective statistics and data science collaboration. Stat labs have reported training 439

graduate (or postgraduate) students (40.8% female) and 912 undergraduate students (41.7% female). From 2014 to December 2021, the total number of LISA 2020 trainees was 2,182.

These 2,182 collaborative statisticians and data scientists have worked on more than 2,274 projects (1,274 projects reported since 2019). Stat lab projects are varied in nature; generally, these projects occur whenever a domain expert collaborates with stat lab staff for statistical advice analysis, or other statistical or data support. These projects may be funded or not; they may require only one meeting, or they may result in a long-term commitment on the part of the stat lab. These projects have directly resulted in a total of at least 134 peer-reviewed publications that include stat lab staff members as authors, as well as 13 non–peer-reviewed publications.

Education and training of non-statisticians take place in several forms. Short courses are brief (less than 1 day); 1-day workshops and multi-day workshops are longer and more involved. All of these have the goal of teaching general statistical and data science principles and techniques to nonstatisticians. Topics could include software applications, specific statistics and data science methods, specialized methods required by domain experts working in specific areas, or anything else in the realm of statistics and data science that benefits the lab's nonstatistician stakeholders. Between 2019 and 2021, 36 stat labs have reported hosting 110 short courses for 2,535 total attendees (31.7% female), 108 1-day workshops for 4,539 total attendees (39.2% female), and 132 multiday workshops for 6,129 total attendees (45.4% female).

2.4 The Purpose of LISA 2020

The original purpose of the LISA 2020 program was to build statistics and data science infrastructure in developing countries by helping to create stat labs to train students and staff to become effective interdisciplinary collaborators; collaborate with researchers, businesses, and policymakers to enable and accelerate research and data-based decisions; and teach short courses and workshops to improve statistical skills and data literacy widely (Vance 2015a). As LISA 2020 has grown and succeeded in achieving its original goal of creating a network of at least 20 stat labs by the year 2020, its purpose has evolved to focus more on the impacts it can facilitate based on the statistics and data science capacity for developing its constituent stat labs have already created.

According to Jacob and Meek, "Leading scientists can play an important bridging role in leveraging global science for local development and serve as conduits between the local and global. But they need support and training, particularly with respect to research management and leadership, if they are to effectively play such bridging roles" (2013, p. 343). The purpose of LISA 2020 is to provide such support and training. Specifically, three intertwined purposes or objectives have emerged for the LISA 2020 program as shown in Figure 2.5:

1. To facilitate and expedite individual stat labs' growth along the path from potential labs to new labs to strong and sustainable labs that serve as mentors to other labs
2. To enable and accelerate stat labs' transforming of evidence into action (TEA) for development
3. To form a community of stat labs to publicize their successes and advocate for their extraordinary potential.

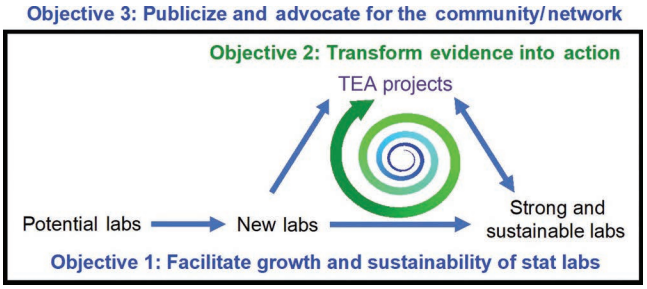


FIGURE 2.5
Three objectives of the LISA 2020 program.

2.4.1 Why LISA 2020 Pursues These Objectives

Stat labs are the foundation of the LISA 2020 program. We believe that the core idea of the stat lab, as described in Vance and Pruitt (2022), is so powerful in advancing the missions of universities for the benefit of society that all universities should have one. But creating a new stat lab is not enough. New stat labs must build their own capacity to become strong and sustainable, and these strong, well-established labs must mentor other labs on their own progression toward sustainability. As a network of stat labs in various stages across this spectrum, LISA 2020 is well positioned to facilitate and expedite individual stat labs' development.

Our vision is for collaborative statisticians and data scientists to collaborate with data producers and data decision-makers to transform evidence (data) into action for development so that more local communities, countries, and regions can benefit from the extraordinary power of statistics and data science to positively impact society. Statistics and data science can be a powerful tool for social impact (Augustin et al. 2021), but statisticians and data scientists must intentionally focus on the outcome of a project, i.e., the *action* for development, rather than on just the production of statistical findings, conclusions, or recommendations. This requires a shift in mindset (Vance and Love 2021) that can be facilitated by the LISA 2020 Network. By intentionally focusing on the *action* for development resulting from the project, stat labs can tailor their statistics and data science work. In many cases, a simpler (though still correct) statistical analysis can be more effective in producing action than a complicated one because it is more understandable and more easily implemented (Stallings 2014).

Since the concept of the stat lab that uses projects to train its students and build its own capacity is still relatively new—having been introduced in the literature by Vance (2015b) not many years ago—a community is needed to support this new concept and to develop and share best practices. Being part of an organization larger than itself can help stat labs become strong and sustainable. As Vance and Pruitt (2022) write, connections with lab mentors and the LISA 2020 Network provide encouragement, support, and guidance to stat labs as they encounter both challenges and triumphs. Stat labs can share their experiences, lessons learned, and best practices to strengthen other labs and the whole network.

Ultimately, LISA 2020 is pursuing these purposes because of the need to build statistics and data science capacity in developing countries (Love et al. 2022). Statistics is widely taught at the university level, and we believe that the application of statistical knowledge has become imperative in virtually all areas of human endeavors because decisions in many areas of modern societies should be based on the collection and analysis of empirical data. Yet, without the proper application of statistical methods, the risk of improper

or inefficient data collection increases, as does the risk that the analysis of research data provides suboptimal results, eventually leading to wrong decisions (Awe and Vance 2014; Jerven 2013). It is also necessary for statisticians to possess effective collaboration skills so as to be able to communicate technical knowledge effectively (Vance and Smith 2019). Hence, the importance of stat labs as an aid in teaching and learning statistics in the 21st century cannot be over-emphasized (Vance 2015b).

2.4.2 How LISA 2020 Pursues These Objectives

We believe that many important technical and nontechnical skills are better learned early in one's career as a statistician at the undergraduate and graduate levels in the university (Awe et al. 2015). Collaborative statisticians at LISA 2020 labs in developing countries are being trained to follow Doug Zahn's POWER structure for organizing and facilitating collaboration meetings (Zahn 2022) and create shared understanding (Vance et al. 2022). Two features of this structure are the wanted conversation and the time conversation. The wanted conversation establishes what the participants want to get out of the meeting and the time conversation establishes how long the meeting will last. We have found that when working with American and Brazilian researchers, these techniques add helpful structure to the meetings (Vivacqua et al. 2018).

The LISA 2020 Secretariat, which was funded by USAID from 2018 to 2022, has put in place policies and procedures to help potential labs advance along the path to becoming newly established and operational (see www.lisa2020.org/about/processes-and-policies). These policies and procedures provide a structure that has been successfully effectuated by many stat labs (Vance and Pruitt 2022). They provide a blueprint for stat labs to build a foundation and facilitate and expedite stat labs' growth toward becoming strong and sustainable labs. The primary way, however, that LISA 2020 accomplishes its three purposes is by developing and sharing best practices throughout the network via various communication channels, special network events and committees, and exchanges and site visits.

Communication: LISA 2020 publishes a weekly newsletter to highlight stat lab activities, promote their successes and triumphs, and share best practices to overcome common challenges (see www.lisa2020.org/newsletter). The network also has a WhatsApp group for members to do the same in a less formal manner. Twice per month, the LISA 2020 Network convenes via Zoom to discuss important topics, see presentations of the experiences of other labs, network at a personal level via informal conversation in small breakout rooms, and share best practices. A newly established lab may bring a question to the Zoom meeting that is answered by more experienced labs that had the same challenge and were able to overcome it. Sharing these lessons learned helps to strengthen all labs and form a robust network.

Special network events and committees: LISA 2020 has convened three symposia, the most recent being an online symposium comprised of 12 sessions on Zoom from October 2020 to May 2021 on the overall theme, "Connecting to Build Capacity to Transform Evidence into Action & Celebrate the International Year of Women in Statistics and Data Science." In conjunction with the online plenary symposium events, four stat labs organized local symposium events and multiple stat labs organized five regional events with financial support from the ISI and their World Bank Statistics Capacity Trust Fund. These regional events occurred in Ghana, Northern Nigeria, Tanzania, Southeast Nigeria, and Southwest Nigeria. Both the local and regional events provided opportunities to promote the successes of the stat labs, share best practices, and spur collaborations through in-person networking.

Another notable special activity of LISA 2020 is the development of a certification program by a network committee to both reward and incentivize students, interns, and faculty/staff to work in a stat lab. The program will have local and global levels of certification so that a student, for example, working in one of the labs will first earn the local certificate, and as she gains more experience, can then be nominated to earn the global LISA 2020 Certificate. The certification program is currently in development and will be comprised of trainings in and assessments of technical skills, collaboration skills, and experience collaborating with domain experts to transform evidence into action. The certification program was identified by the network as a major need to help standardize operations across stat labs, improve quality, and overcome the challenge of having too few volunteer statistics and data science collaborators working in the labs.

Exchanges and site visits: We have found that the best way to improve the quality of stat labs, share best practices, and create a robust network is to encourage exchanges between lab staff and site visits to observe other labs in practice. UI-LISA has operated an internship program providing 3 or more months of industrial experience for students to work in the lab since 2015 (Adepoju et al. 2022). Initially, just for its own students, this program has expanded to accept students from other universities who return to their university fully equipped to contribute to their own newly established lab. The COVID-19 pandemic has curtailed much travel and exchange opportunities between stat labs. We hope to resume exchanges and site visits in the coming years and to secure the funding necessary to facilitate them, especially "South-South" exchanges between the stat labs.

2.5 LISA 2020 as a Big Tent for Collaborative Statistics and Data Science

An analogy we have found useful for explaining the purpose of the LISA 2020 program is the "Big Tent." As the ASA's 107th president, Dr. Bob Rodriguez advocated for the ASA to become the "Big Tent" for statistics (Rodriguez 2013). Rodriguez explained, "Big tents do three things. They attract all kinds of people, they serve them on the inside, and they are highly visible on the outside" (2013, p. 2). LISA 2020 can become the Big Tent for collaborative statistics and data science by creating a community of stat labs to share best practices, welcoming statisticians and data scientists to create new stat labs; serving the stat labs within the network through trainings, mentorship, certificates, symposia, exchanges, etc. so that they can become strong and sustainable; enabling and accelerating their transformation of evidence into action (TEA); making the successes of stat labs highly visible to outside observers; and advocating for their extraordinary potential (see Figure 2.6).

By becoming the Big Tent for collaborative statistics and data science, the LISA 2020 program can achieve its purpose in attracting new stat labs, serving to build their capacity to become strong and sustainable labs that transform evidence into action for development, and publicizing their work to make visible the extraordinary impacts of statistics and data science collaboration.

Already, the LISA 2020 program has made the following impacts (Vance and Pruitt 2022):

- Improving statistical literacy among students and researchers in developing countries

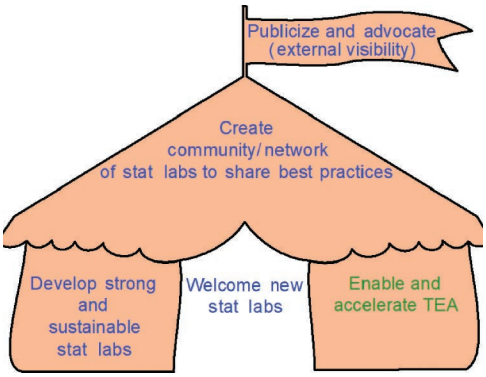


FIGURE 2.6
LISA 2020 can become the Big Tent for statistics and data science collaboration.

- Accelerating research development by offering expert statistical advice to researchers in the institutions
- Aiding graduate students to complete their research appropriately and graduate on time
- Helping to increase the number of Master's and Ph.D. candidates being produced
- Aiding researchers to publish quality papers in high impact journals
- Analyzing data for various research grants, funding proposals, and projects
- Educating graduate students and researchers through regular short courses on modern statistical packages and software
- Causing increased enrollment of students into the various statistics programs in the LISA 2020 Network
- Encouraging collaboration between statisticians and researchers from other disciplines (i.e., domain experts)
- Helping researchers to keeping abreast with current research trends and best practices.

2.6 Conclusion

In this chapter, we explained the history, key events, current state, and future directions of the LISA 2020 program to become the Big Tent for statistics and data science collaboration. With roots as far back as 1948, LISA 2020 began in 2012 and achieved its initial goal of creating a network of at least 20 stat labs in developing countries by the year 2020. Currently comprised of 35 full member stat labs in 10 countries and 13 additional proposed or transitional stat labs, the LISA 2020 Network is well positioned to deliver on its current purposes to help create new stat labs, build their capacity to become strong and sustainable, enable and accelerate their transforming of evidence into action for development, and publicize their inevitable successes in applying statistics and data science to positively impact humanity.

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